ACADEMIC RESEARCH POSTER

Authors: Tiane Souto Couto, Keegan Cullen, Damian French, Aidan Hughes, Joshua Roberts, Lucas Robertson

Affiliations: The University of Manchester

01
ABSTRACT AND INTRODUCTION

Filler

02
THEORY

The Lighthill-Whitham_Richards (LWR) models traffic flow in a 1D description, similar to the analysis of an compressible fluid, dictated by mass conservation. Traffic is treated as having a macroscpic density ρ and flow rate q , governed by the core principles of conservation laws and the flow-density relation. The former can be derived directly from the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho u) = 0$$

over a 1D road segment. Imposing vehicles to act as a mass and treating density as vehicles per unit length yields

$$\frac{\partial \rho}{\partial t} + \frac{\partial q}{\partial x} = 0$$

the conservation law disctating the LWR model. However, this equation contains unknowns, making it undetermined. A relationship used between q and ρ here is known as the fundamental diagram. Acting as as the experimentally obtained relation of traffic, this shows increasing flow with density to a critical point; after which congestion occurs causing reduced flow. The Greenshields model assumes a linear speed-density relation, yielding the fundamental diagram

$$q = \rho u = v_{max}\rho(1 - \frac{\rho}{\rho_{max}})$$

.

Applying this with the conservation condition gives a PDE describing characteristic wavespeed

$$\frac{\partial q}{\partial \rho} = v_{max}(1 - \frac{2\rho}{\rho_{max}})$$

highlighting the pertubations that propogate at speeds that depend on a local density. This forms shock waves that act as density discontinuities; defining the LWR model as a whole.

With the waves characteristics descibed by the LWR-Greenshields model, it becomes posible to determine how traffic density evolves with time. Prediciting the behaviour and estimaring travel time between two given points requires a numerical solution of the PDE shown above. The Cell Transition Model (CTM) provides a framework using discrete cells, meaning real traffic data can be applied to the LWR model, acting as a numerical solution to the characteristic wave behaviour.

04
RESULTS

What are the outcomes of your research? Are you able to support your hypothesis or reject your hypothesis? This section will explain your findings.

05
DISCUSSION

Use this section to give a visual representation of your research. Use bullet points, graphs, tables, charts, or graphics that support your study.

- First supporting statement
- Second supporting statement
- Third supporting statement

06
CONCLUSION

The conclusion should summarize 2–3 key findings from your research. Highlight the most significant results and explain their importance to the research field. You may also include potential future directions the research can explore.

References: